Problem Set VIII, Macroeconomics III
University of Copenhagen

Department of Economics, University of Copenhagen

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Exercise 1

The goal of this exercise is to familiarize yourself with the Diamond-Mortensen-Pissarides framework from the lecture notes, but also understand its versatility. While in the lecture, it was applied to the labor market, this exercise applies it to the rental market in Copenhagen. In the model, nobody can buy an apartment, renting is the only option.

Some share of individuals in the city are living with their parents, looking to rent their own apartment (share h, "houseless share"), the rest are renting ($r = 1 - h$, "renting share"). Each month, a fraction δ of all renters leave their apartment, while a fraction f of all searchers find one (assume f and δ are constant for now). The framework is static, hence you will not need time-indices.

(a) Write down an expression for the share of agents looking for housing, as a function of δ and f. If f rises, does the share of renters rise or fall?

Now let's focus on the landlords. For simplicity, they just earn the rent money m from their tenants. They discount the future at rate β. Hence, the total discounted utility of a landlord who has a tenant is

$$L(m) = m + \beta\left[(1-\delta)L(m) + \delta V\right]$$

If the landlord loses her current tenant, she will need to find a new one. V is the present discounted utility of landlords who have an apartment to sublet but no tenant. Finding a tenant entails cleaning the apartment, taking nice pictures, getting the utilities checked, and posting an ad on bolig.dk. All these costs are summarized as κ. We assume that having an empty apartment is not costly in itself. (This is a strong assumption, apartments deteriorate even if nobody lives in them, but it makes the math easier).

(b) Provide an expression for the present discounted utility of a landlord without a tenant (a "vacancy"), as a function of κ, η (the probability of finding a tenant), $L(m)$ and V. Do not impose free entry yet. Keep in mind that landlords discount the future at rate β.*(Hint: This expression will be very similar to the one in the lecture notes.)*

Now, we endogenize the apartment finding rate $f(\theta)$ and the apartment filling rate $\eta(\theta)$. They now depend on the tightness of the rental market, $\theta = v/h$, the number of apartments on offer relative to the number of searchers. Assume there is some matching function $M(\theta)$ that microfounds these expressions. The function $f(\theta)$ is increasing in θ, $\eta(\theta)$ decreases in θ. Both lie between 0 and 1, as they are probabilities.

(c) Assume there is free entry for landlords. What does this imply for the value of posting a vacant apartment?

(d) Using your answer to question (b), show that $L(m) = \frac{\kappa}{\beta\eta(\theta)}$ if entry is free.

(e) Derive the break-even rent m as a function of rental market tightness. *Hint: you have two expressions for $L(m)$.*

The renters who have not found their own apartment live at home and get some utility b. When they find their own apartments, they are happier, with higher utility q, but they have to pay rent. Hence, the utility of a searcher and a renter are

$$H = b + \beta\left[f(\theta)R(m) + (1 - f(\theta))H\right]$$
$$R(m) = q - m + \beta\left[(1-\delta)R(m) + \delta H\right]$$

(f) Derive an expression for the renters surplus S_R. To an individual, what is the value of finding an apartment? All else equal, how is it affected if b rises, i.e., living at home becomes more attractive?

When renters and landlords meet, they bargain over the rent m, according to Nash bargaining. They split up the total surplus from a match ($\Omega = S_R + L(m)$) according to their bargaining power. The bargaining power of the renters is ξ. This leads to

$$m = -\xi\kappa\theta + (1-\xi)(q-b)$$

You do not need to show this.

(g) Explain the economic intuition behind this equation.

(h) Write down the equation that defines the general equilibrium level of the rental market tightness θ^*. You do not need to solve it. (You cannot, since no function form for $\eta(\theta)$ was provided.) Draw a diagram that clarifies the relationship between θ and w.

(i) What happens to rental rates m if it becomes more attractive to live at home? Explain using the diagram from the figure above.

(j) What happens to rental rates m if agents in the model become more patient, i.e., β increases?